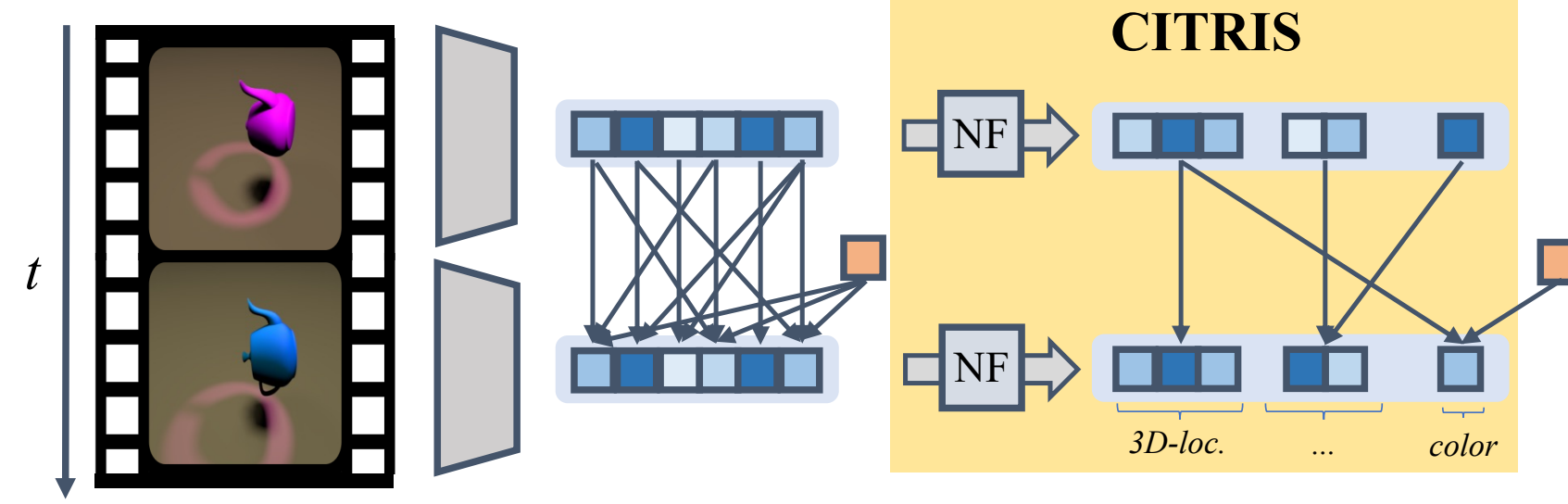




Problem statement

- Causal representation learning considers the identifiability of causal factors from high-dimensional observations such as images for causal reasoning
- We consider causal factors that are temporally related, and have possibly been intervened upon between time steps
- The causal factors may be multidimensional, e.g. position in 3D or rotation in multiple axis. Up to what degree can we yet identify the factors?



Causal Identifiability from Temporal Intervened Sequences

- The general setup in terms of a causal graph is shown on the right
- C_i^t - a causal factor at time step t , I_i^{t+1} - observed intervention target, X^t - high-dimensional observation (e.g. image), R^{t+1} - confounder among interventions
- Key aspect: each causal variable is causally related to only one intervention target
- Main theoretical result: we can identify the **minimal causal variables**, i.e. the information/mechanism of a causal variable which strictly depends on the interventions

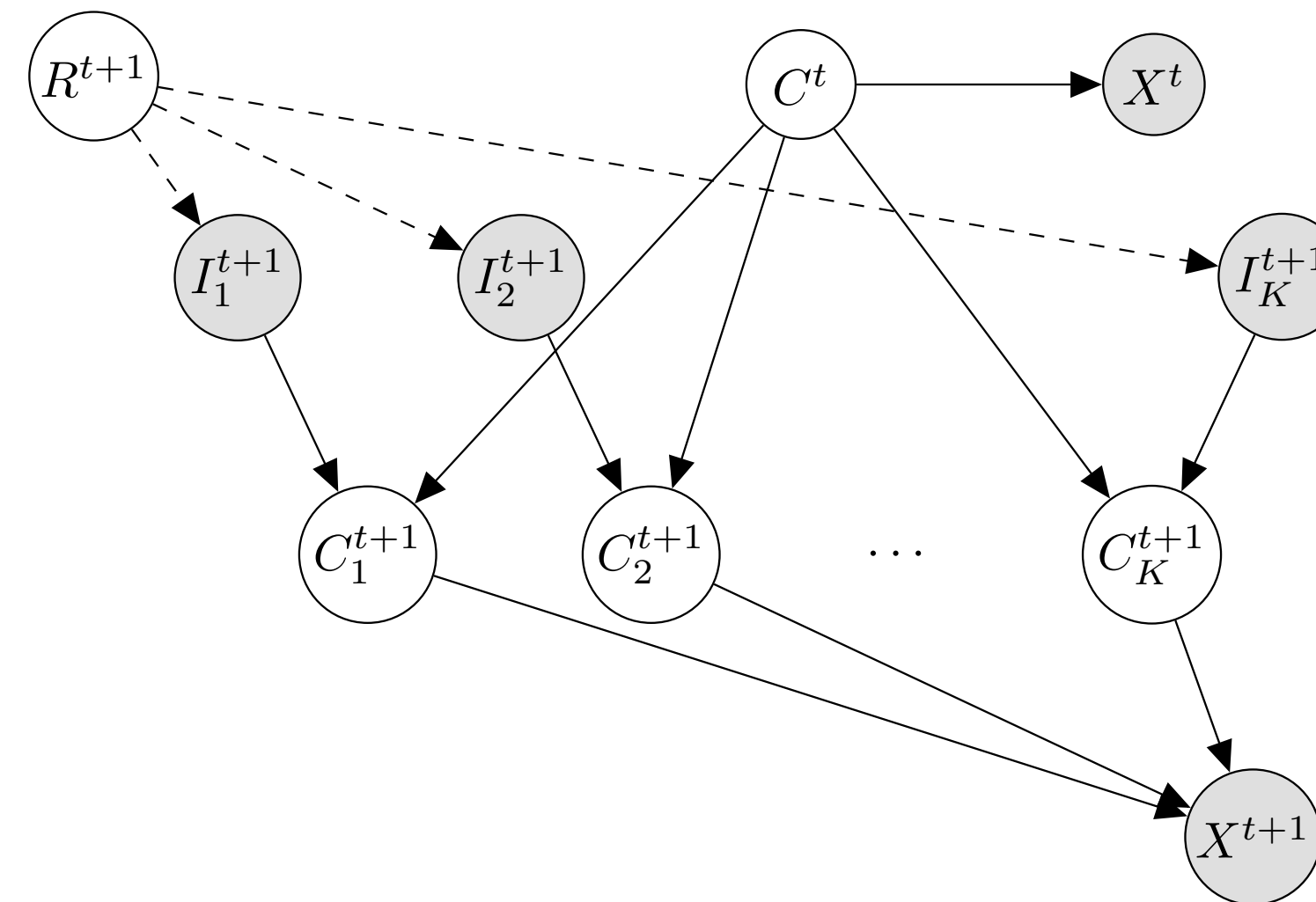
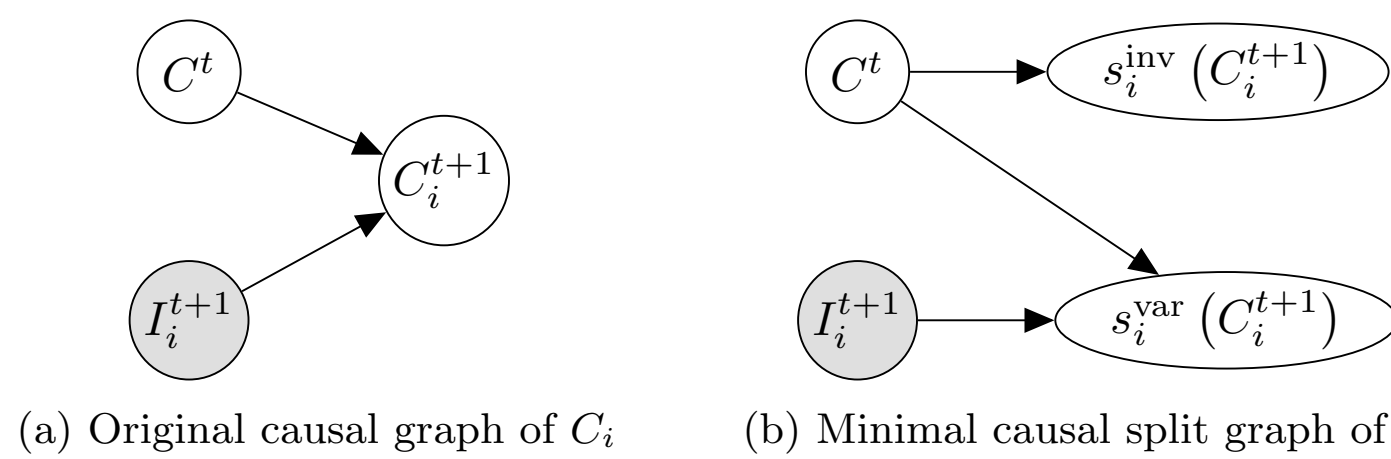
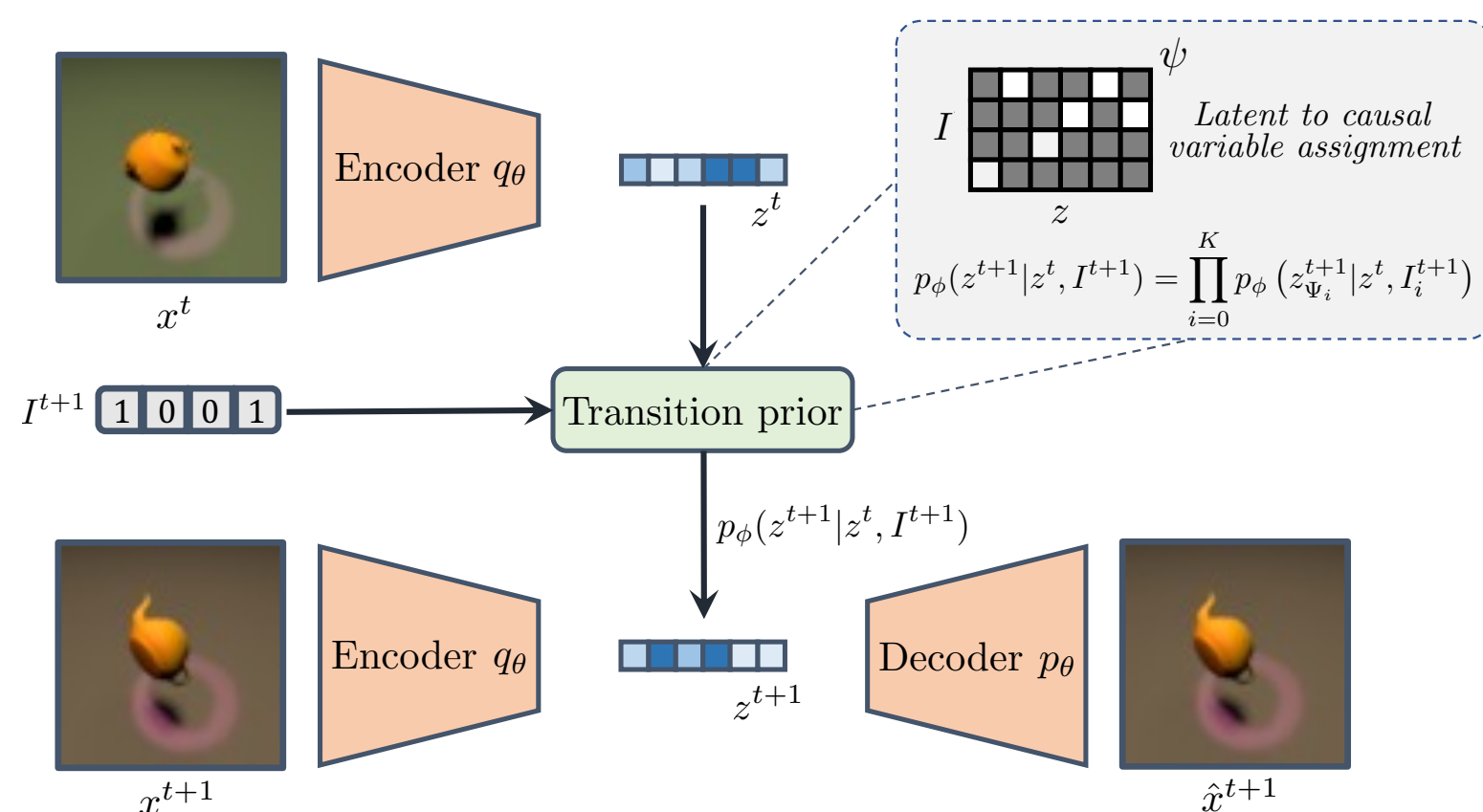


Figure: The split into minimal causal variables visualized as a causal graph

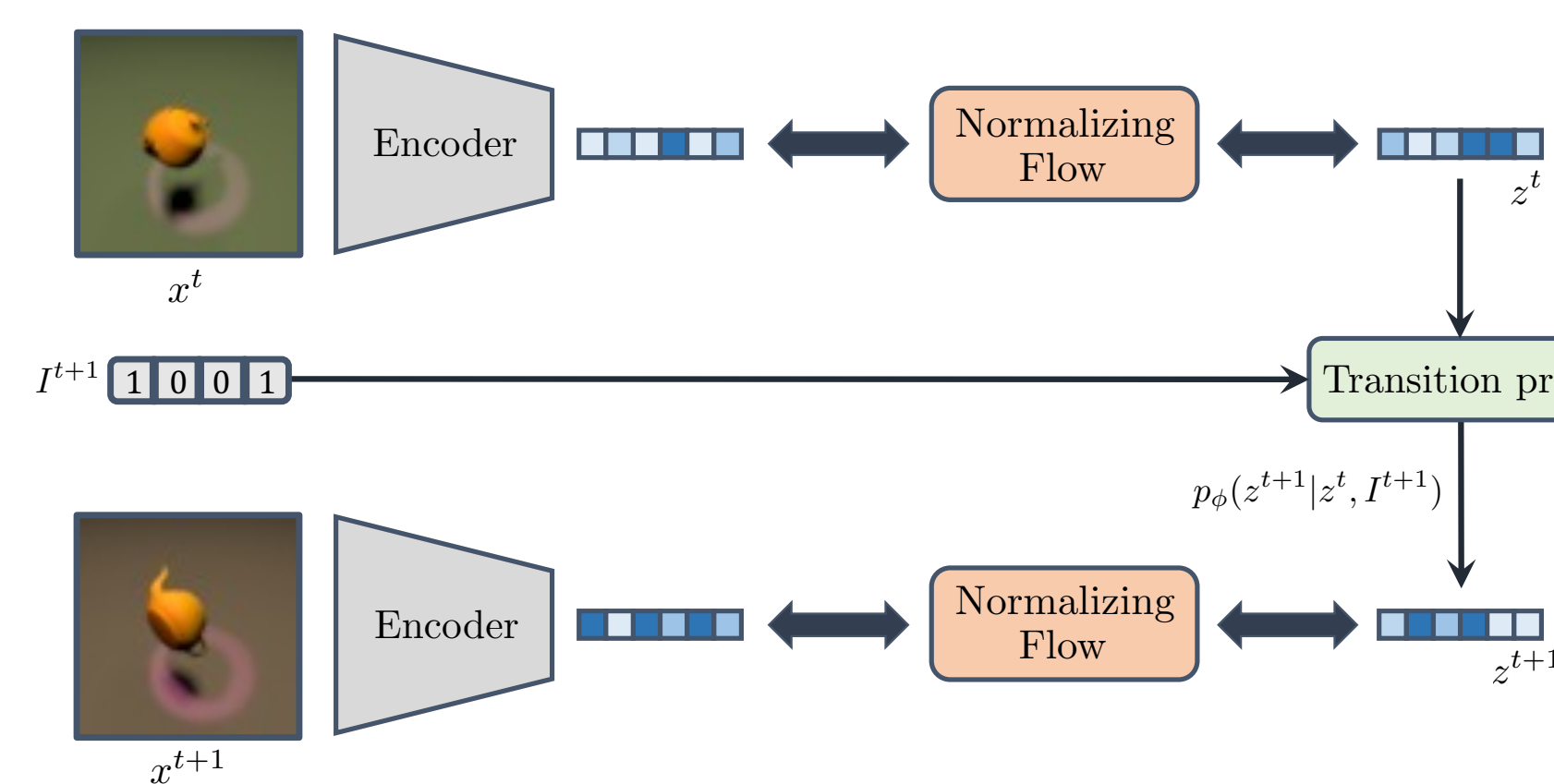
CITRIS-VAE

- Variational Autoencoder structure with specific latent prior structure
- Learn assignment of latent dimension $\rightarrow$ causal factors via Gumbel Softmax
- The prior of each latent conditioned on only its respective intervention target
- Target classifier: additional network that tries to predict intervention targets from latent space – guiding optimization and disentanglement



CITRIS-NF

- Instead of VAE, use pretrained autoencoders and normalizing flows
- NF learns a map between entangled autoencoder representation and disentangled, causal representation
- AE only trained on observational, single-image data; NF can generalize beyond seen interventional data



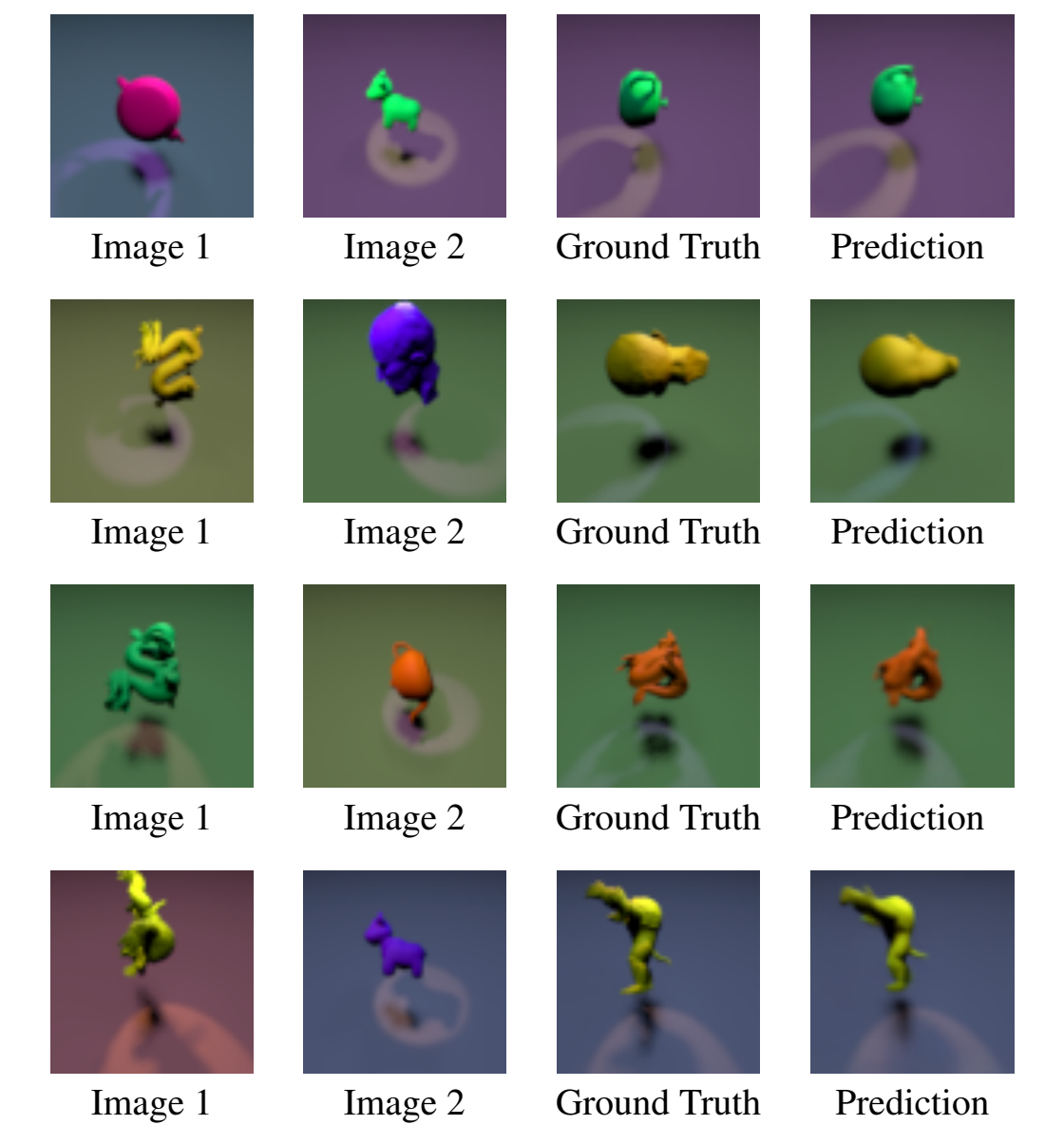
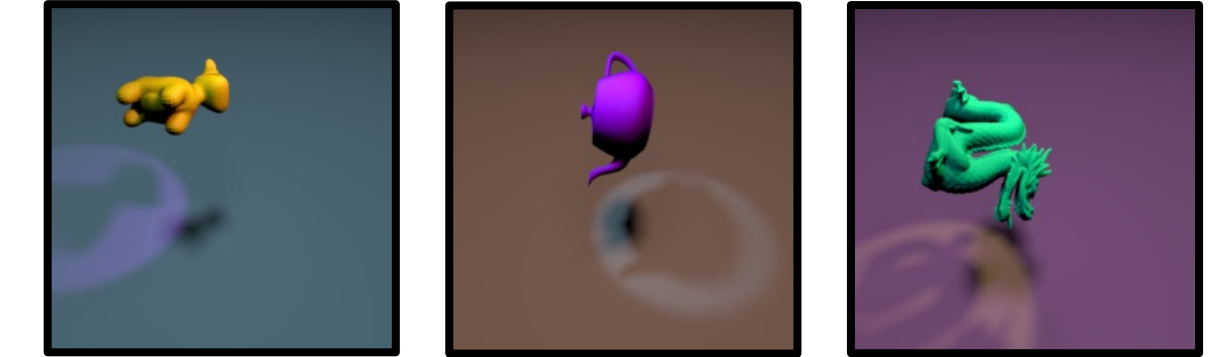
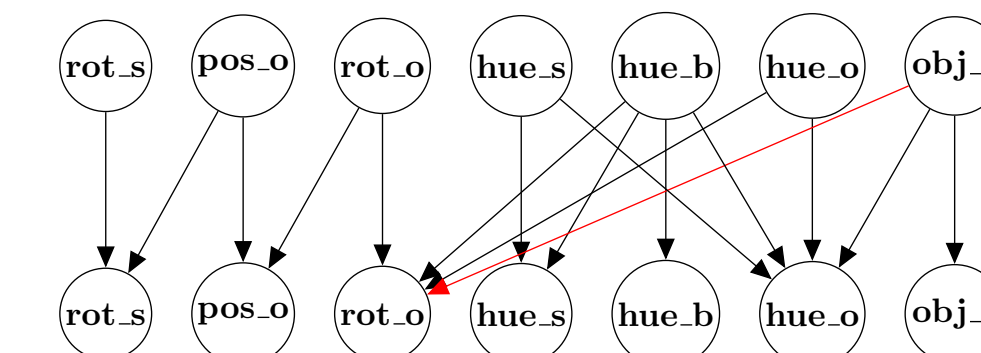
Experiments

- Experiments on videos with interventions

Temporal Causal3DIdent

- 3D rendered images with 7 causal factors: 3D position, 2D angles, spotlight rotation, spotlight hue, background hue, object hue, object shape
- Evaluation metrics: (1) correlation between latents and causal factors; (2) novel combinations of causal factors by performing combination in latent space
- CITRIS outperforms baselines and AE+NF combination works very well
- Generalization experiments: train NF on 5 object shapes, check its disentanglement on 2 unseen shapes – still works reasonable well

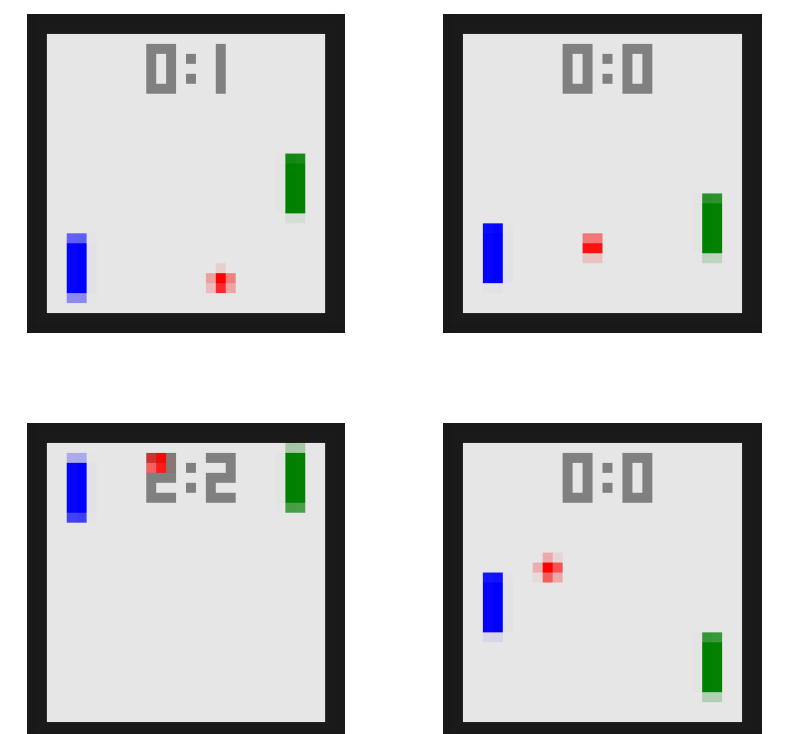
	Triplet dist $\downarrow$		Correlation metrics			
	Mean	R^2 diag $\uparrow$	R^2 sep $\downarrow$	Spearman diag $\uparrow$	Spearman sep $\downarrow$	
SlowVAE	0.40	0.61	0.23	0.59	0.27	
iVAE*	0.27	0.80	0.29	0.77	0.28	
CITRIS-VAE	0.21	0.89	0.10	0.88	0.12	
CITRIS-NF	0.07	0.98	0.04	0.97	0.08	
- 5 shapes	0.09	0.98	0.05	0.97	0.10	
- 2 unseen shapes	0.23	0.94	0.15	0.93	0.19	



Interventional Pong

- Reinforcement Learning inspired environment – Pong with observed interventions
- Common game dynamics of Pong with soft interventions on paddles as moving up-down, ball interventions as changing position/vel., no interventions on score
- CITRIS is able to disentangle the causal variables, and maps the score into separate variables for intervention-invariant information

	Triplets $\downarrow$	R^2 diag $\uparrow$	R^2 sep $\downarrow$	Spearman diag $\uparrow$	Spearman sep $\downarrow$
SlowVAE	0.34	0.61	0.17	0.66	0.23
iVAE*	0.09	0.91	0.04	0.92	0.06
CITRIS-VAE	0.03	0.99	0.01	0.99	0.05
CITRIS-NF	0.02	1.00	0.04	1.00	0.10



Takeaways

- We can identify multidimensional causal variables from high-dimensional, temporal data up to their minimal causal variable
- CITRIS disentangles factors by enforcing structure in the prior and learns alignment of latent to causal variables
- CITRIS-NF generalizes to unseen instances of causal factors, holding promise for future work

Check out our paper and code for details!